Mobile Cellular Communications (5G)

PART I

Introduction to cellular networks

I. Objectives

The objectives for this practical work are:

- Get familiar with common networking analyzer tools
- Apply those tools Cellular technologies
- Make an initial technical characterization of Cellular technologies
- Have a first contact with cellular communications concepts

II. Duration

This work is expected to take 30 minutes.

III. Material

This work will use:

1. Own students Android/iOS mobile terminals (smartphones and tablets)
2. Available public cellular networks in the laboratory location
3. Commercial cellular networks from operators
4. Free analyzer tools, required to be previously installed at own students mobile terminals:
    1. CellMapper

IV. Procedures

1) Cell Mapper

Reference figures for the steps bellow:

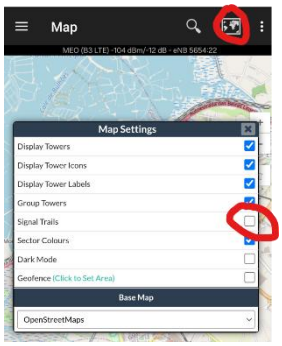


Fig 1.a

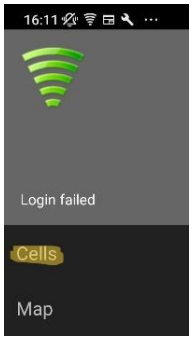


Fig 1.b

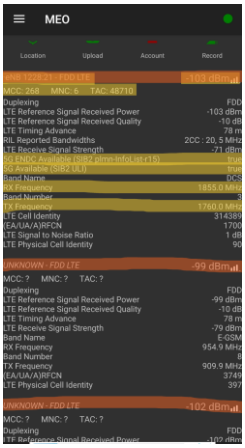


Fig 1.c



Fig 1.d



Fig 1.e

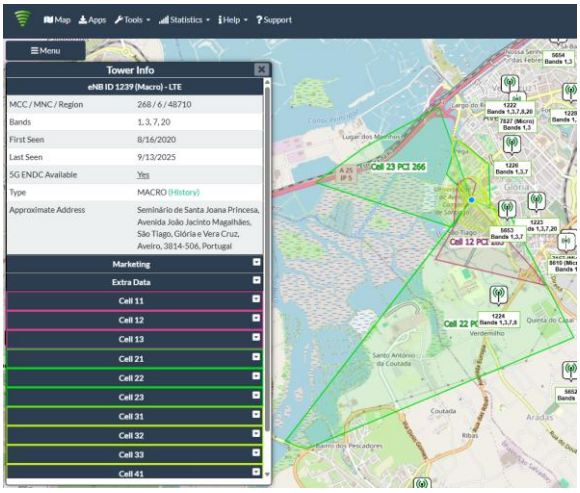


Fig 1.f



Fig 1.g

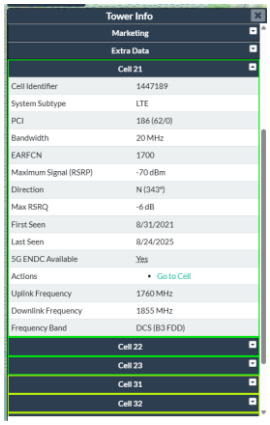


Fig 1.h

1) **Enable location** in your Android mobile terminal
2) **Start** *CellMapper* tool
3) Go to the 'Map Settings' icon in the top right corner and **disable 'Map Trails'** for better visibility in the following steps (Fig 1.a)
4) **Access** the menu (top left corner) and select '**Cells**' (Fig 1.b)
    - In the main screen you will get a list of available cells in your vicinity (Fig 1.c), with the first being the one your terminal is connected to
5) **Check** your network operator name in the top left corner
6) **Note**, for your case, the value highlighted in yellow/orange ('eNB xxxx:yy', 1228:21 in the example); **identify** the used technology (2/3/4/5G)
7) **Note** the MCC and MNC values; **compare** with colleagues in different operators
8) **Scroll** down in the main screen and check the existence of other cells being sensed by your terminal; **identify** the parameter and the respective value by which they are ordered; maybe the first value (your cell) does not present the highest value. **Explain that**.
9) **Analyse** other presented values and **compare** with other cells in the vicinity
10) **Access** the menu (top left corner) and select '**Map**'

11) **Press** the search icon in bottom right corner of the screen (Fig 1.d) to 'Select Provider'; this shows the operators available in your location; **alternatively** go to the search icon in the top (which allows you to select from all the operators in the world) and, in the 'Provider' field, start writing 'Portugal', until you see listed the Portuguese public operators **and select your operator ('Vodafone'/'MEO'/'NOS'/'Digi'); select '4G – LTE' and 'All' Bands**; after that, you will see in the screen only cell towers belonging to that operator
12) **Zoom out** to a level allowing you to see from Aveiro to Tomar (*if you zoom out too much, you will get an error message regarding the number of requests to the DB*); **wait a few seconds** and see how numbers reflecting the number of deployed cells start to appear; **are you surprised by the numbers?** you may zoom in and out in several country zones
13) Change to another operator and **try an estimation on the total number of existing cell towers** in Portugal
14) **Press** the arrow icon in bottom right corner to center the map in your location
15) **Zoom/pan** until you see a blue line showing the radio connection between your terminal and the cell tower it is connected to (see Fig 1.g)
16) **Select** that tower (a 'Tower Info' popup window will appear; see the example in Fig 1.f, for MEO, 4G and the Tower close to Reitoria/Seminário); some polygons; in the popup window, see the provided information:
    - **MCC and MNC**
    - **Bands** (see URL in references to understand this)
17) As seen in the example Fig 1.f, there are several 'Cells' under the same 'Tower'; **Scroll down** until you see the cell with the number observed in step 6 (yy or 21, in the example); this is the cell in the Tower your terminal is connected to; check the following values:
    - **Bandwidth**
    - **Direction** (this is the same as azimuth)
    - **Maximum RSRP and maximum RSRQ**
    - **Up and downlink frequencies**
    - **Frequency Band**
18) Try to find that Cell in the polygons that appeared when the Tower was selected or just press 'Go to cell' and it is highlighted in the map; compare its average orientation with the 'Direction' information shown above.
19) Go again to the **search icon** in top of the screen and **select '5G – NR'** as 'Network' and return to the map; you may have to select another operator (not all are shown)
20) **Select** one of the shown cell towers and **check the gNB ID**
21) **Scroll down** until you find a cell in Frequency band n78; **compare** the indicated bands, bandwidth, up and downlink frequencies with the previously obtained 4G information; check the FDD and TDD indication in both cases
22) Go to other cells and find other bands used by 5G NR
23) Now change to 2G, select one of the towers, check its **BTS ID** and operating bands
24) Now, change to 3G. is there anything 'strange'?

V. Usefull links

Android Apps (Google Play)

https://play.google.com/store/apps/details?id=cellmapper.net. 0

https://play.google.com/store/apps/details?id=com.signalmonitoring.wifimonitoring

https://play.google.com/store/apps/details?id=org.zwanoo.android.speedtest

Cellular

https://en.wikipedia.org/wiki/LTE_frequency_bands

https://en.wikipedia.org/wiki/5G_NR_frequency_bands

https://www.cellmapper.net

Other

https://www.wilsonamplifiers.com/blog/finding-cell-tower-locations-the-complete-guide/

https://www.m2catalyst.com/apps/network-cell-info/faq

PART II

5G networks

VI. Objectives

The objectives of this laboratory are:

- Identify configuration parameters required for the different components
- Understand the main procedures in a mobile cellular (5G) at the control and data planes by running opensource implementations of the main components

VII. Duration

This laboratory should last 4h30, divided in 3 classes.

VIII. Used tools

This laboratory will use:

a) An opensource implementation of a 5G Core: Free5GC [free5GC] [free5gcwiki]

b) A gNB and UE opensource implementation: UERANSIM [ueransim]

c) A VirtualBox VM with both components already installed in the laboratory PCs

d) Wireshark also installed in the laboratory PCs

The VM is also available via SSH at port 2222 for user '***ubuntu***' (e.g. '***ssh -p 222 ubuntu@localhost***', from the hosting machine); password is '***ubuntu***' for users '***ubuntu***' and ***root***.

IX. Network diagram

1. 5G Core components (provided by Free5GC) are represented by light blue (control plane) and purple boxes (UPF, dataplane), gNBs by ligth orange boxes, and UEs by green boxes
2. With UERANSIM, the 5G-NR radio interface ('Radio Link', RL) is emulated over UDP between the UEs (11, 12 and 21, green boxes) and the gNBs (gNB1 and gNB2, light orange boxes) they are connected to.
3. IP addresses:
   a. 10.0.123.0/24 (red): SBI; Core components, Web Console and DB (control plane)
   b. 10.0.124.0/24 (orange): N2 interfaces (control plane)
   c. 10.0.130.0/24 (purple): N3 interfaces (data plane)
   d. 10.0.140.0/24 (dark red): N4 interfaces (control plane)
   e. 10.0.20[1|2].0/24 (green): radio interfaces emulation
   f. 10.1.[1|2].0/24 (blue): N6 DNNs (data plane)
4. Via the 'Host' entity, emulated UEs can reach the Internet.
5. A ***hosts*** file has been added to Wireshark (***/root/.config/wireshark***) for IP addresses resolution so that Wireshark presents components' names instead of IP addresses, allowing a faster and better interpretation of the messages exchange (see that file contents in Annex F at the end).

6. The shown MongoDB in the diagram component serves as persistent data repository for the other components while the network is running (you will see a significant volume of messages with it).

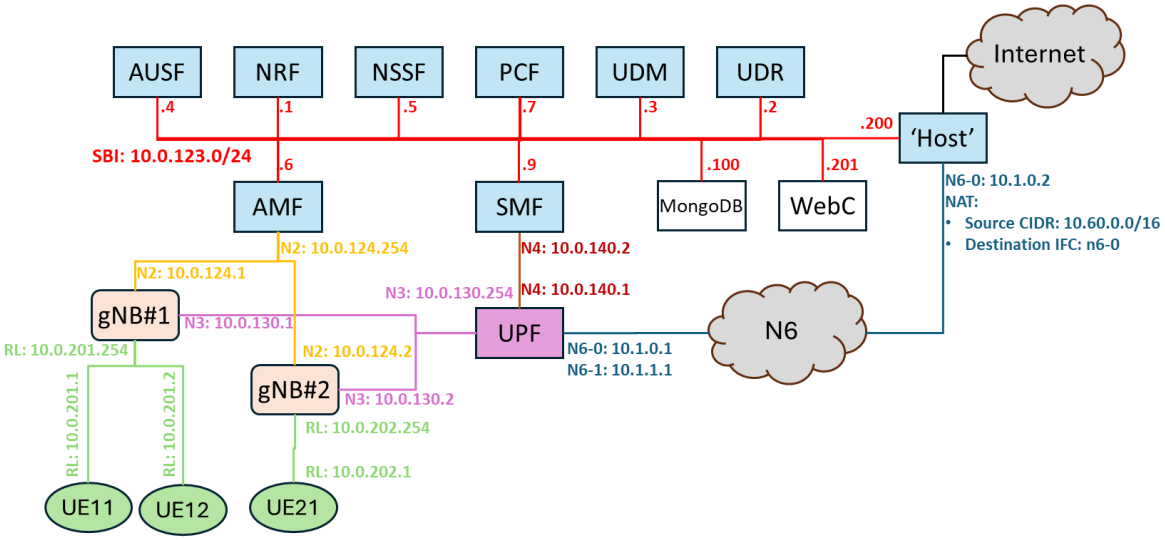


Figure 1: Network diagram

X. Procedures

1. Introduction

During the laboratory execution, based on the provided VirtualBox Virtual Machine, the 5G network components will be started and stopped, following this order (numbers in parentheses indicate the guide execution step):

1. **5G Network: (a) 5G Core (3.7) → (b) gNB1 (4.3) → (c) gNB2 (4.5) →**
2. **UEs creation: → (d) UE provisioning at the 5G Core (5.2) →**
3. **UEs start: → (e) UE11 (5.4) → (f) UE12 (6.5) → (g) UE21 (6.7) →**
4. **Stop the system: → (h) UEs, gNBs and 5G Core (9.1 and 9.3)**

Linux *Namespaces* are used to run each of the nine 5GC Network Functions (AMF, AUSF, NRF, NSSF, PCF, SMF, UDM, UDR, UPF) inside its own namespace [konrad]. This allows the usage of Wireshark (**shall be started with 'sudo'**) to capture traffic packets exchanged between any two NFs, on their own interfaces (you will get the list of interfaces present in the following screen capture, when selecting capturing interfaces in Wireshark, after 5G Core components have been started).

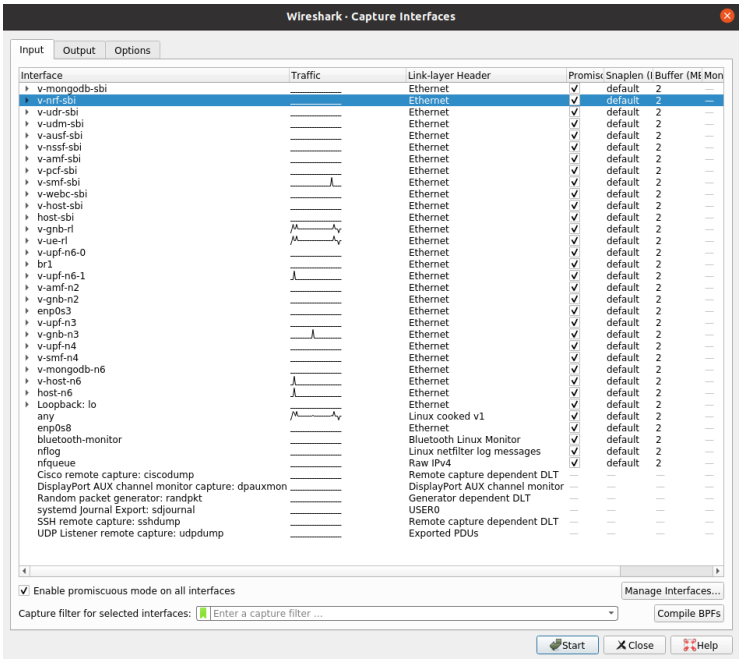


Figure 2: Logical interfaces as seen in Wireshark, after 5G Core start

2. Configurations analysis

1) Analyse the yaml configuration files in the list below (2.1.a), located in folder *~/5GLab/netns5g/config* (you may open them with the File Manager) and search for the listed configuration parameters in 2.1.b.

   a. **Files**:

      i. 5G Core: **amfcfg**.yaml, **smfcfg**.yaml, **upfcfg**.yaml

      ii. 5G RAN: free5gc-**gnb1**.yaml, free5gc-**gnb2**.yaml

      iii. 5G UEs:

      - free5gc-**ue11**.yaml
      - free5gc-**ue12**-**sl1**.yaml, free5gc-**ue12**-**sl2**.yaml
      - free5gc-**ue21**.yaml

```
ubuntu@ubuntu-VirtualBox:~/5GLab/netns5g/config$ more amfcfg.yaml
info:
  version: 1.0.3
  description: AMF initial local configuration

configuration:
  amfName: AMF # the name of this AMF
  ngapIpList:  # the IP list of N2 interfaces on this AMF
    - 10.0.124.254
  sbi: # Service-based interface information
    scheme: http # the protocol for sbi (http or https)
    registerIPv4: 10.0.123.6 # IP used to register to NRF
    bindingIPv4: 10.0.123.6  # IP used to bind the service
    port: 8000 # port used to bind the service
    tls: # the local path of TLS key
      pem: config/TLS/amf.pem # AMF TLS Certificate
      key: config/TLS/amf.key # AMF TLS Private key
  serviceNameList: # the SBI services provided by this AMF, refer to TS 29.518
    - namf-comm # Namf_Communication service
    - namf-evts # Namf_EventExposure service
    - namf-mt   # Namf_MT service
    - namf-loc  # Namf_Location service
    - namf-oam  # OAM service
  servedGuamiList: # Guami (Globally Unique AMF ID) list supported by this AMF
    # <GUAMI> = <MCC><MNC><AMF ID>
    - plmnId: # Public Land Mobile Network ID, <PLMN ID> = <MCC><MNC>
        mcc: 001 # Mobile Country Code (3 digits string, digit: 0~9)
        mnc: 01 # Mobile Network Code (2 or 3 digits string, digit: 0~9)
      amfId: cafe00 # AMF identifier (3 bytes hex string, range: 000000~FFFFFF)
  supportTaiList:  # the TAI (Tracking Area Identifier) list supported by this AMF
    - plmnId: # Public Land Mobile Network ID, <PLMN ID> = <MCC><MNC>
```

Figure 3: Example of 5G Core entity (AMF) configuration file (partial)

```
ubuntu@ubuntu-VirtualBox:~/5GLab/netns5g/config$ more free5gc-gnb1.yaml
mcc: '001'          # Mobile Country Code value
mnc: '01'           # Mobile Network Code value (2 or 3 digits)

nci: '0x000000010'  # NR Cell Identity (36-bit)
idLength: 32        # NR gNB ID length in bits [22...32]
tac: 1              # Tracking Area Code

linkIp: 10.0.201.254   # gNB's local IP address for Radio Link Simulation (Usually same with local IP)
ngapIp: 10.0.124.1   # gNB's local IP address for N2 Interface (Usually same with local IP)
gtpIp: 10.0.130.1    # gNB's local IP address for N3 Interface (Usually same with local IP)

# List of AMF address information
amfConfigs:
  - address: 10.0.124.254
    port: 38412

# List of supported S-NSSAIs by this gNB
slices:
  - sst: 0x1
    sd: 0x010203
  - sst: 0x2
    sd: 0x112233

# Indicates whether or not SCTP stream number errors should be ignored.
ignoreStreamIds: true
```

Figure 4 Example of 5G RAN (gNB) configuration file (partial)

```
ubuntu@ubuntu-VirtualBox:~/5GLab/netns5g/config$ more free5gc-ue11.yaml
# IMSI number of the UE. IMSI = [MCC|MNC|MSISDN] (In total 15 digits)
supi: 'imsi-001010000000011'
# Mobile Country Code value of HPLMN
mcc: '001'
# Mobile Network Code value of HPLMN (2 or 3 digits)
mnc: '01'

# Permanent subscription key
key: '8baf473f2f8fd09487cccbd7097c6862'
# Operator code (OP or OPC) of the UE
op: '8e27b6af0e692e750f32667a3b14605d'
# This value specifies the OP type and it can be either 'OP' or 'OPC'
opType: 'OPC'
# Authentication Management Field (AMF) value
amf: '8000'
# IMEI number of the device. It is used if no SUPI is provided
imei: '356938035643811'
# IMEISV number of the device. It is used if no SUPI and IMEI is provided
imeiSv: '4370816125816111'

# List of gNB IP addresses for Radio Link Simulation
gnbSearchList:
  - 10.0.201.254

# UAC Access Identities Configuration
uacAic:
  mps: false
  mcs: false

# UAC Access Control Class
uacAcc:
  normalClass: 0
  class11: false
  class12: false
  class13: false
  class14: false
  class15: false

# Initial PDU sessions to be established
sessions:
  - type: 'IPv4'
    apn: 'internet'
    slice:
      sst: 0x1
      sd: 0x010203
```

Figure 5: Example of 5G User Equipment configuration file (partial)

b. **In those files, search for and identify the following parameters**:

   i. MCC and MNC

   ii. NR Cell Identities and TACs

   iii. Supported slices at gNB1 and gNB2 (SST+SD)

   iv. Supported DNN

   v. List of SUPIs (UE11, UE12 and UE21)

c. **For entities in the SBI (e.g. AMF) check other parameters and information**:

   i. IP addresses matching with the figure above

   ii. Exposed services (e.g. Namf***)

3. 5GC start

2) Open a terminal window
3) Change to directory (~/5GLab/netns5g) containing the scripts needed to setup and run the 5G environment
4) Initialize environment (create the namespaces and the virtual interfaces)

   ~/5GLab/netns5G$ ***sudo ./5Gsetup.sh***

5) Check created namespaces and connecting links

   ~/5GLab/netns5G$ ***sudo ip netns*** # lists created namespaces

   ~/5GLab/netns5G$ ***sudo ip link*** # lists created links

6) Start a Wireshark

   ~/5GLab/netns5G$ ***sudo wireshark &***

```
ubuntu@ubuntu-VirtualBox:~/5GLab/netns5g$ sudo ./5Gsetup.sh
net.ipv4.ip_forward = 1
create namespace mongodb
create_interface: if_local mongodb-sbi, if_bridge v-mongodb-sbi, namespace mongodb
configure_interface mongodb-sbi v-mongodb-sbi mongodb 10.0.123.100
disableIpv6IfRequested: interface mongodb-sbi, namespace mongodb
create namespace nrf
create_interface: if_local nrf-sbi, if_bridge v-nrf-sbi, namespace nrf
configure_interface nrf-sbi v-nrf-sbi nrf 10.0.123.1
disableIpv6IfRequested: interface nrf-sbi, namespace nrf
create namespace udr
create_interface: if_local udr-sbi, if_bridge v-udr-sbi, namespace udr
configure_interface udr-sbi v-udr-sbi udr 10.0.123.2
disableIpv6IfRequested: interface udr-sbi, namespace udr
create namespace udm
create_interface: if_local udm-sbi, if_bridge v-udm-sbi, namespace udm
configure_interface udm-sbi v-udm-sbi udm 10.0.123.3
disableIpv6IfRequested: interface udm-sbi, namespace udm
create namespace ausf
create_interface: if_local ausf-sbi, if_bridge v-ausf-sbi, namespace ausf
configure_interface ausf-sbi v-ausf-sbi ausf 10.0.123.4
disableIpv6IfRequested: interface ausf-sbi, namespace ausf
create namespace nssf
create_interface: if_local nssf-sbi, if_bridge v-nssf-sbi, namespace nssf
configure_interface nssf-sbi v-nssf-sbi nssf 10.0.123.5
disableIpv6IfRequested: interface nssf-sbi, namespace nssf
create namespace amf
create_interface: if_local amf-sbi, if_bridge v-amf-sbi, namespace amf
configure_interface amf-sbi v-amf-sbi amf 10.0.123.6
disableIpv6IfRequested: interface amf-sbi, namespace amf
create_interface: if_local amf-n2, if_bridge v-amf-n2, namespace amf
configure_interface amf-n2 v-amf-n2 amf 10.0.124.254
disableIpv6IfRequested: interface amf-n2, namespace amf
```

Figure 6: start of namespaces (not complete)

7) Start the capture in the interface 'br1' (this will capture all the traffic; you can start other Wireshark instances at specific interfaces, e.g. 'v-amf-sbi' to see only that specific traffic)
8) Start the 5G Core (free5gc)

   ~/5GLab/netns5G$ ***sudo ./5Gstart.sh***

   Check there are no errors reported in shown messages; if no, check for a potential solution in Annex G

   When seeing 'all services are running', 5G Core Network Functions (NFs) have started, each in its own namespace.

   a. With 'ps -gaxu | grep free5gc' check the different running processes and the order they started (see the processes ID)

```
2024-10-28T22:24:18Z [INFO][LIB][PFCP] Remove Request Transaction [1]
2024-10-28T22:24:18Z [INFO][SMF][App] Received PFCP Association Setup Accepted Response from UPF[10.0.140.1]
deploying service: [namespace: webc, path: ../free5gc/webconsole/bin/webconsole, args: -c ./config/webuicfg.yaml -p ../free5gc/webconsole/public -l ./logs/webconsole.log -lc ./logs/free5gc.log]
2024-10-28T22:24:21Z [INFO][WebUI][Init] WebUI Log level is set to [trace] level
2024-10-28T22:24:21Z [INFO][WebUI][App] webui
2024-10-28T22:24:21Z [INFO][WebUI][App] webconsole version:
        free5GC version: v3.2.1
        build time:      2022-10-26T08:33:42Z
        commit hash:     69864fe6
        commit time:     2022-05-25T05:24:43Z
        go version:      go1.14.4 linux/amd64
2024-10-28T22:24:21Z [INFO][WebUI][Init] Server started
skipping deployment of: host
skipping deployment of: gnb1
skipping deployment of: ue11
skipping deployment of: ue12
skipping deployment of: gnb2
skipping deployment of: ue21
all services running
```

Figure 7: start of 5G Core (last messages)

   b. Observe the script output and register, confirming the order by which 5G Core components have been started. Relate the order they appear with the existing inter dependencies.

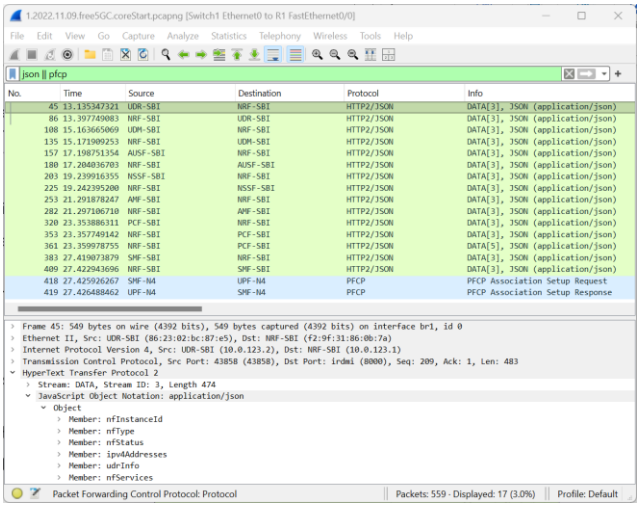
   c. Observe the successive interactions with NRF; what is that for?

9) Stop the capture and identify the involved protocols (to facilitate it, order the capture by the 'Protocol' column by pressing the respective column top); which of those are specific 5G Core protocols?

   a. Complementary, go to 'Statistics' → 'Conversations'; check 'Name resolution' and analyse the results

   b. In addition, go to 'Statistics' → 'Protocol Hierarchy' and identify here the previously identified protocols

10) Identify the dialogs for identified 5G protocols; apply a display filter for those protocols and check the involved entities



    a. Go to 'Statistics' → 'Flow Chart'; observe the sequence of the interactions in the diagram

    b. Filter only by 'json':

        i. Observe the pairs of messages; With which other entity are these talking to? What for?

        ii. UPF does not appear; why?

        iii. Analyse the protocol stack for the exchanged JSON messages

        iv. In one of the exchanged messages, expand the subtrees under JSON; briefly observe the contents of several of these messages

    c. Filter only by 'pfcp':

        i. Which entities are here involved? Observe the message types

        ii. Analyse the protocol stack for PFCP

11) Open and briefly analyse the log files present in ~/5Glab/netns5G/logs for the different entities; SMF example:

```
time="2025-11-04T21:03:25Z" level=info msg="Endpoints: [10.0.130.254]" category=CTX component=SMF
time="2025-11-04T21:03:25Z" level=info msg="Server started" category=Init component=SMF
time="2025-11-04T21:03:25Z" level=info msg="SMF Registration to NRF {071cf68e-68bb-4c6e-b9e1-1060c0b3ad60 SMF REGISTERED 0 0xc00009c2e0 0xc00009c320 [] []  [10.0.123.9] [] <nil> [] [] <nil> 0 0 0 area1 <nil> <nil> <nil> <nil> 0xc0000ae240 <nil> <nil> <nil> <nil> <nil> map[] <nil> false 0xc00009c120 false false []}" category=Init component=SMF
time="2025-11-04T21:03:25Z" level=info msg="Listen on 10.0.140.2:8805" category=PFCP component=SMF
time="2025-11-04T21:03:25Z" level=info msg="Sending PFCP Association Request to UPF[10.0.140.1]" category=App component=SMF
time="2025-11-04T21:03:25Z" level=info msg="Remove Request Transaction [1]\n" category=PFCP component=LIB
time="2025-11-04T21:03:25Z"  level=info  msg="Received PFCP Association Setup Accepted Response from UPF[10.0.140.1]" category=App component=SMF
```

4. Stop and reset the environment

12) **Stop** the UEs, gNB nodes, and the 5G Core in this order (Ctrl-C)

13) **Wait** for final processes to close (this takes some seconds, ending with "NRF terminated")

14) **Delete** the namespaces

    ~/5GLab/netns5G$ **sudo ./5Gcleanup.sh**

15) **Power off** the VM

Annexes

A. 5G System architecture

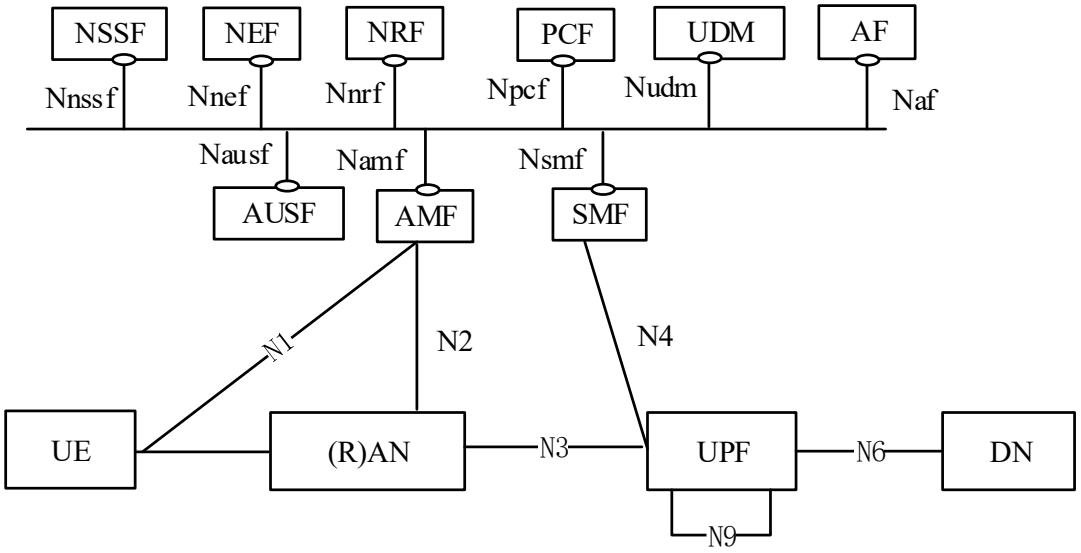


Figure 8: 5G Core, RAN and UE reference system architecture [3gpp]

B. Example procedure

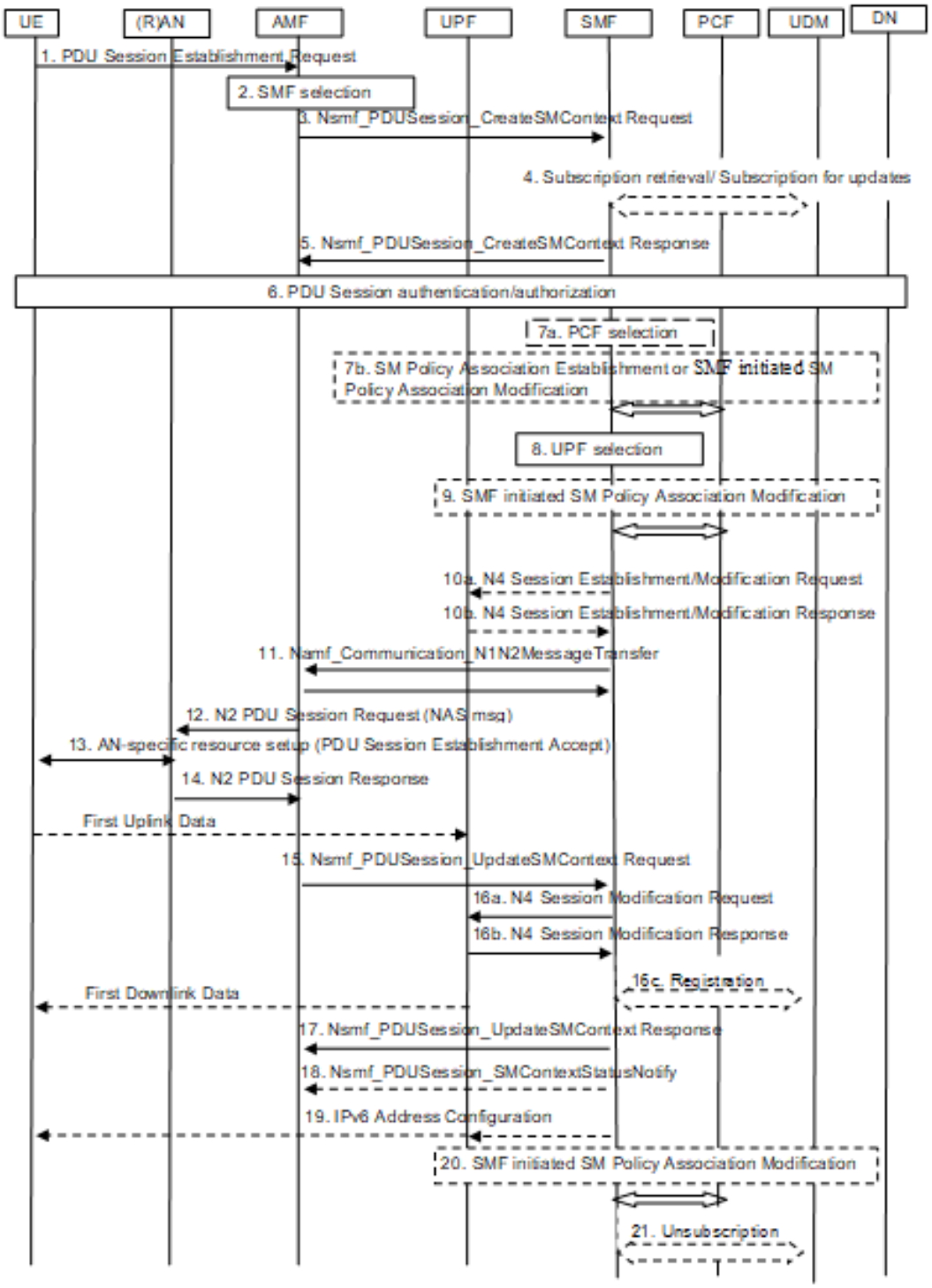


Figure 4.3.2.2.1-1: UE-requested PDU Session Establishment for non-roaming and roaming with local breakout

C. 5G Protocol stacks

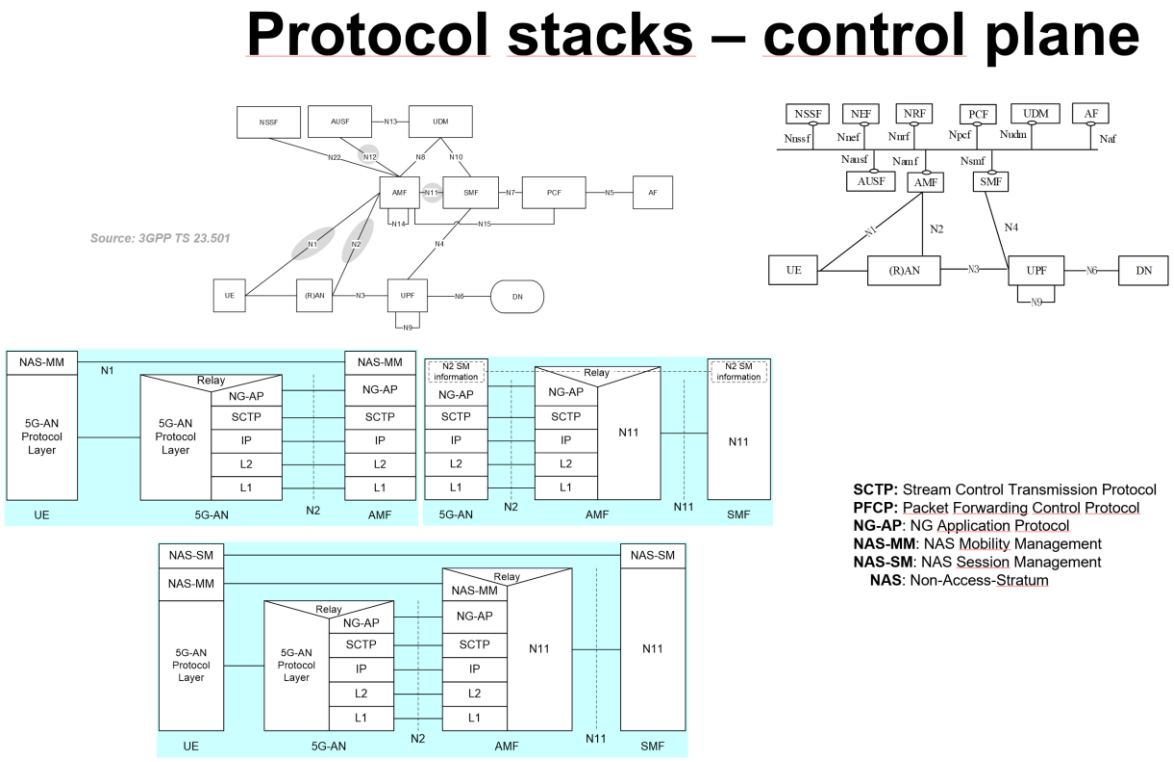


Figure 9: Protocol stacks, control plane

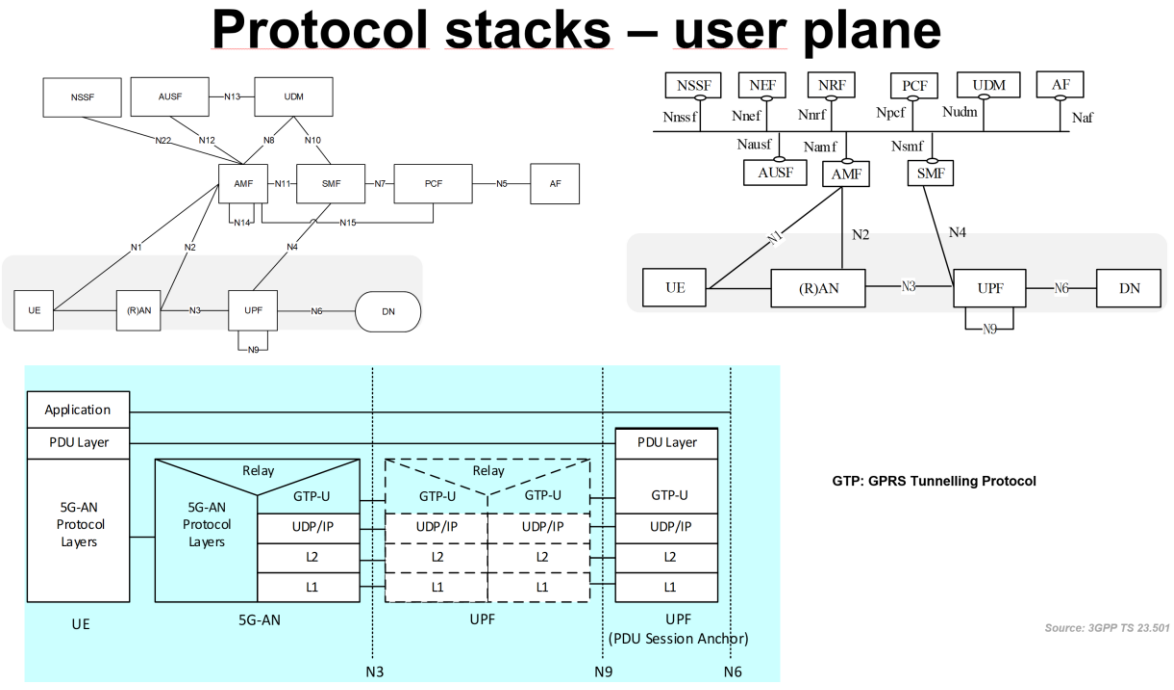


Figure 10: Protocol stacks, user plane

D. Free5GC New Subscriber creation form (UE11)

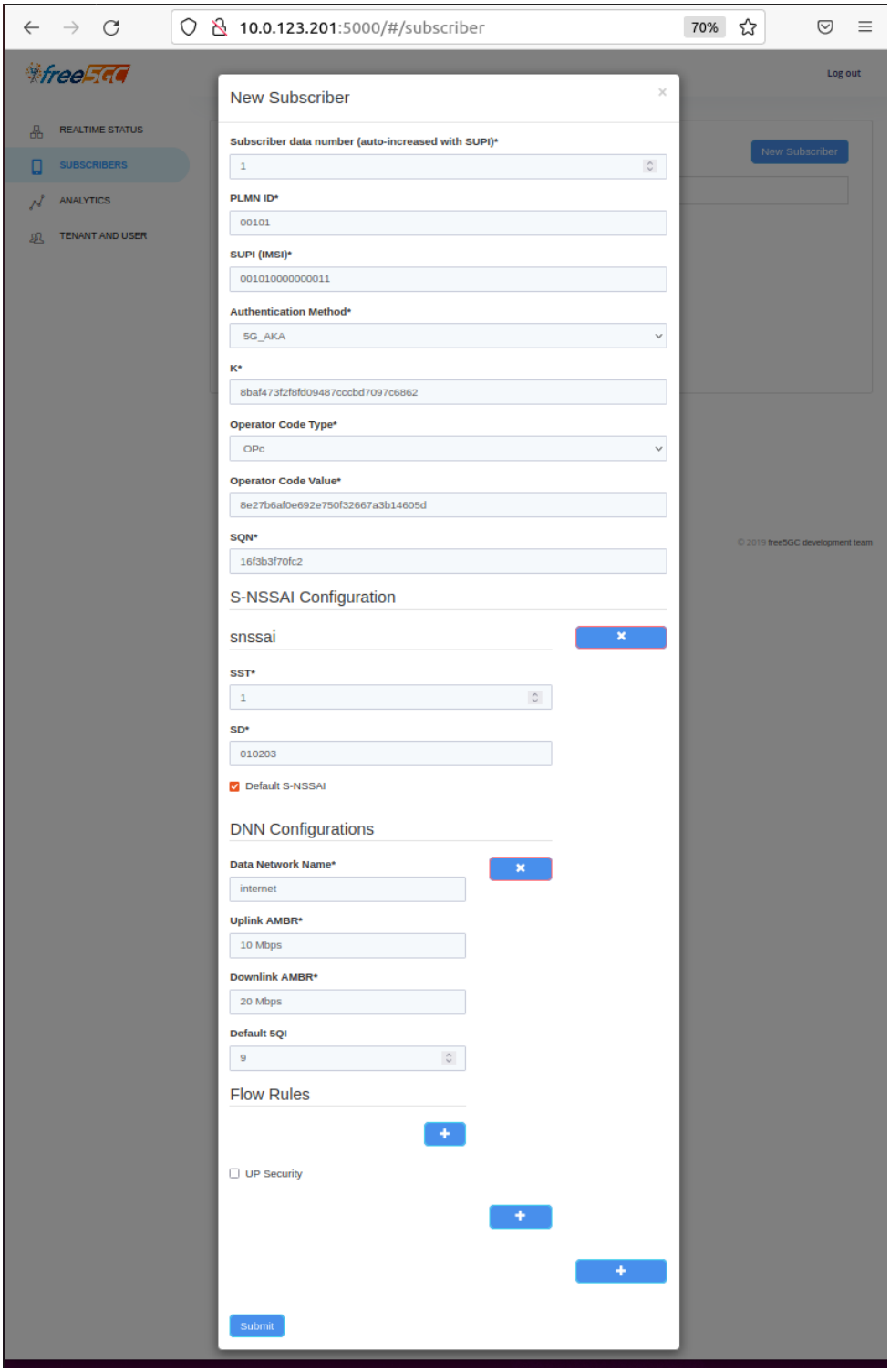


Figure 11: Example of a subscriber provisioning at Free5GC (UE11)

E. Example of a capture with Wireshark (with addresses resolution and display filter)

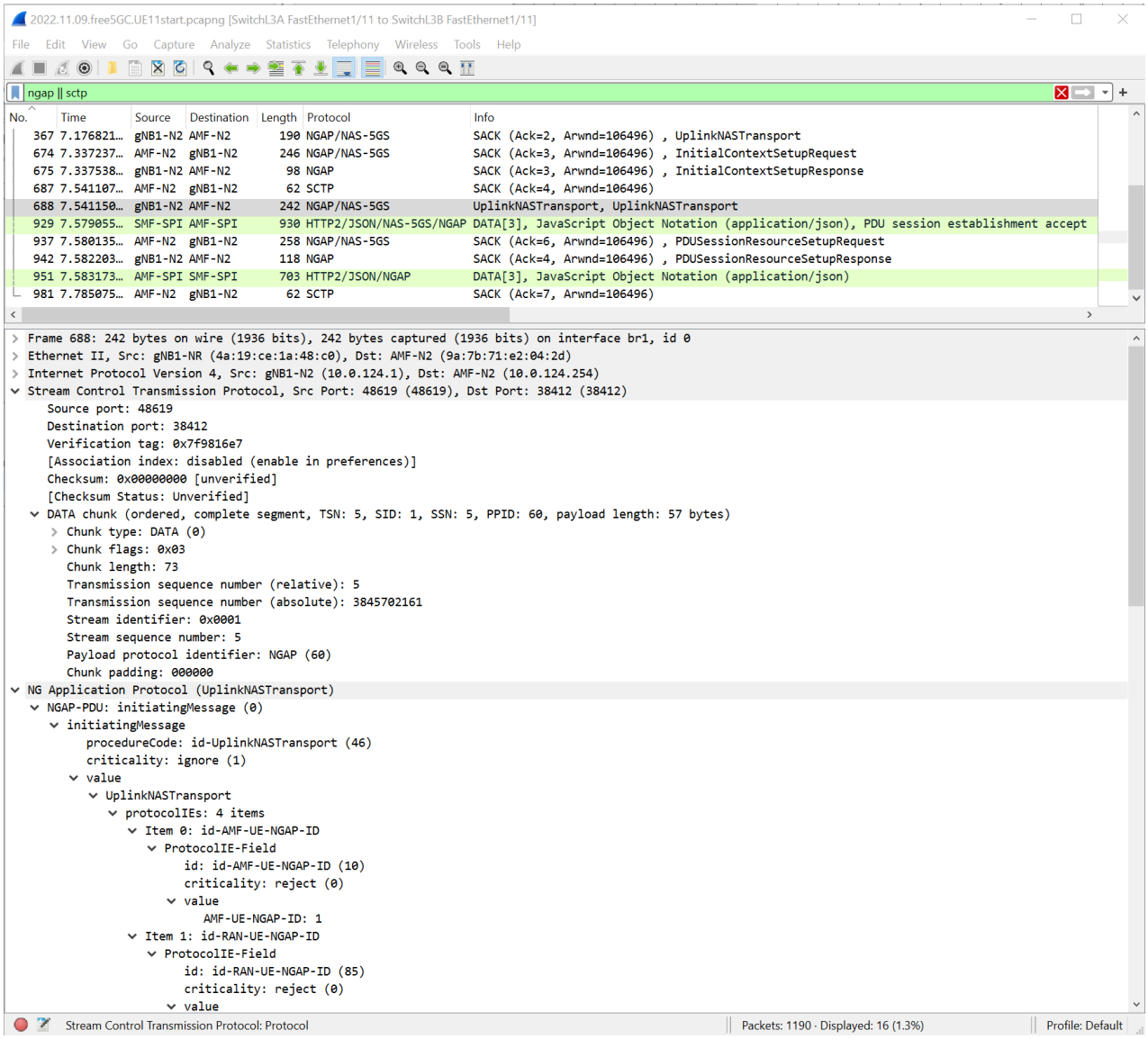


Figure 12: Example of a 5G exchange of messages (gNB – AMF interactions)



Figure 13: Detail of a 5G exchanged message (AMF and gNB via N2 interface)

F. Hosts file

```
#5G Core
10.0.123.1    NRF-SBI
10.0.123.2    UDR-SBI
10.0.123.3    UDM-SBI
10.0.123.4    AUSF-SBI
10.0.123.5    NSSF-SBI
10.0.123.6    AMF-SBI
10.0.123.7    PCF-SBI
10.0.123.9    SMF-SBI
10.0.123.100  MongoDB-SBI
10.0.123.200  hostm
10.0.123.201  WebConsole

10.0.124.254  AMF-N2
10.0.124.1    gNB1-N2
10.0.124.2    gNB2-N2

10.0.140.2    SMF-N4
10.0.140.1    UPF-N4

#5G dataplane
10.1.0.1      UPF-N6
10.1.0.1      Host-N6

#RAN1
10.0.201.1    UE11-NR
10.0.201.2    UE12-NR
10.0.201.254  gNB1-NR

#RAN2
10.0.202.1    UE11-NR
10.0.202.254  gNB1-NR
```

G. GTP (reinstalation)

In the situation at 5GCore start gtp fail with similar errors to these ones:

```
2025-11-04T20:51:00Z [FATA][UPF][Main] panic: runtime error: invalid memory address
or nil pointer dereference

2025-11-04T20:51:11Z  [ERRO][SMF][App]  Failed  to  setup  an  association  with
UPF[10.0.140.1], error:Request Transaction [1]: retry-out
```

1. Stop the 5gCore (Ctrl-C)
2. go to *~/5GLab/gtp5g* and issue the following commands
3. *make clean && make*
4. *sudo make install*
5. restart the 5GCore

H. Useful links

A. Free5GC:

a. [free5Gcore] https://www.free5gc.org/

b. [free5gcwiki] https://github.com/free5gc/free5gc/wiki

c. [konrad] https://github.com/konradkar2/netns5g

B. UERANSIM:

a. [ueransim] https://github.com/aligungr/UERANSIM/wiki

C. 3GPP

a. [3gpp] https://www.3gpp.org